StudyBot: Intelligent Agent with Tools, Reasoning & Memory

Natural Language Processing (AI 2026) • Module 08: Building Chatbots & Agents

Made with GAMMA

What Makes StudyBot Special?

Unlike a plain ChatGPT session, StudyBot combines multiple advanced capabilities into a single intelligent agent that can reason, act, and remember.

Calculator Tool

Performs precise arithmetic with no hallucinated math

Student Lookup

Fetches private grade data an LLM can't know

Memory

Remembers the full conversation across turns

LangGraph ReAct Loop

Reason → Act → Observe → Reason...

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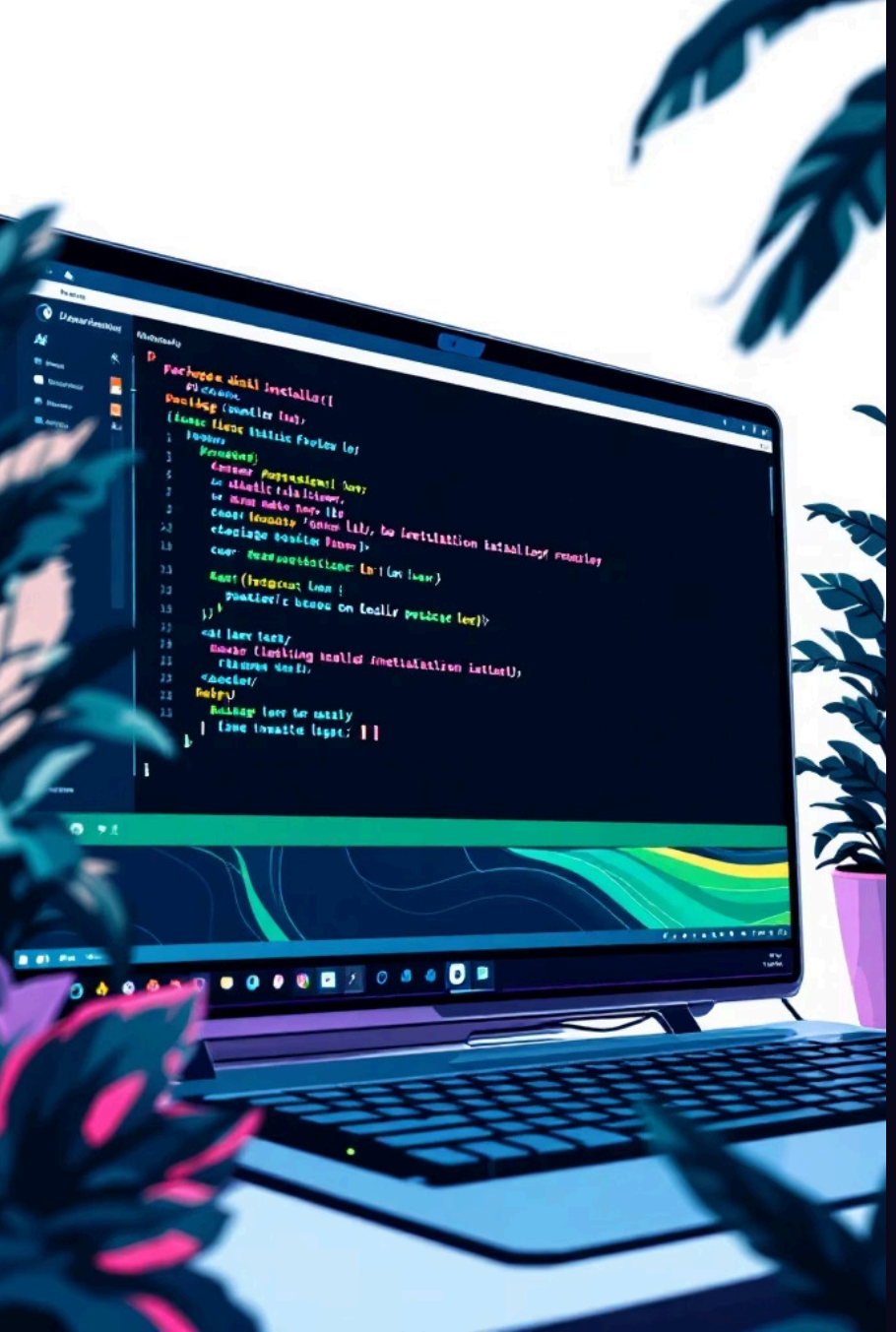

Getting Started: Installation & Setup

Install Dependencies

```
%pip install -q langchain langgraph langchain-ollama
```

Run once to install required packages. You may need to restart the kernel to use updated packages.

Configure Ollama

1. Download from ollama.com and install
2. Run: `ollama pull llama3.2` (~2GB download)
3. Ollama runs at **http://localhost:11434**
4. No API key needed!

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Defining the Tools

Two tools are registered with `@tool` so LangChain exposes their schema to the LLM automatically.

1

Calculator Tool

Evaluates mathematical expressions and returns precise results.
Use for ANY arithmetic, percentages, or grade calculations.

```
calculator(expression: str) -> str
Example: '85 * 0.4 + 90 * 0.6'
```

2

Student Lookup Tool

Retrieves private grade records from the student database.
Returns formatted grade information.

```
student_lookup(student_name: str, subject: str = None) -> str
Example: student_lookup('Alice', 'Math')
```

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Building the LLM & Agent State

01

Initialize ChatOllama

Uses Ollama running locally with llama3.2 model

02

Bind Tools

LLM learns tool schemas automatically

03

Define Agent State

TypedDict with messages using add_messages reducer

04

Set System Prompt

Rules: always use tools, be concise and encouraging

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LangGraph Architecture

LangGraph uses a state machine with two nodes that create the ReAct loop:

Agent Node

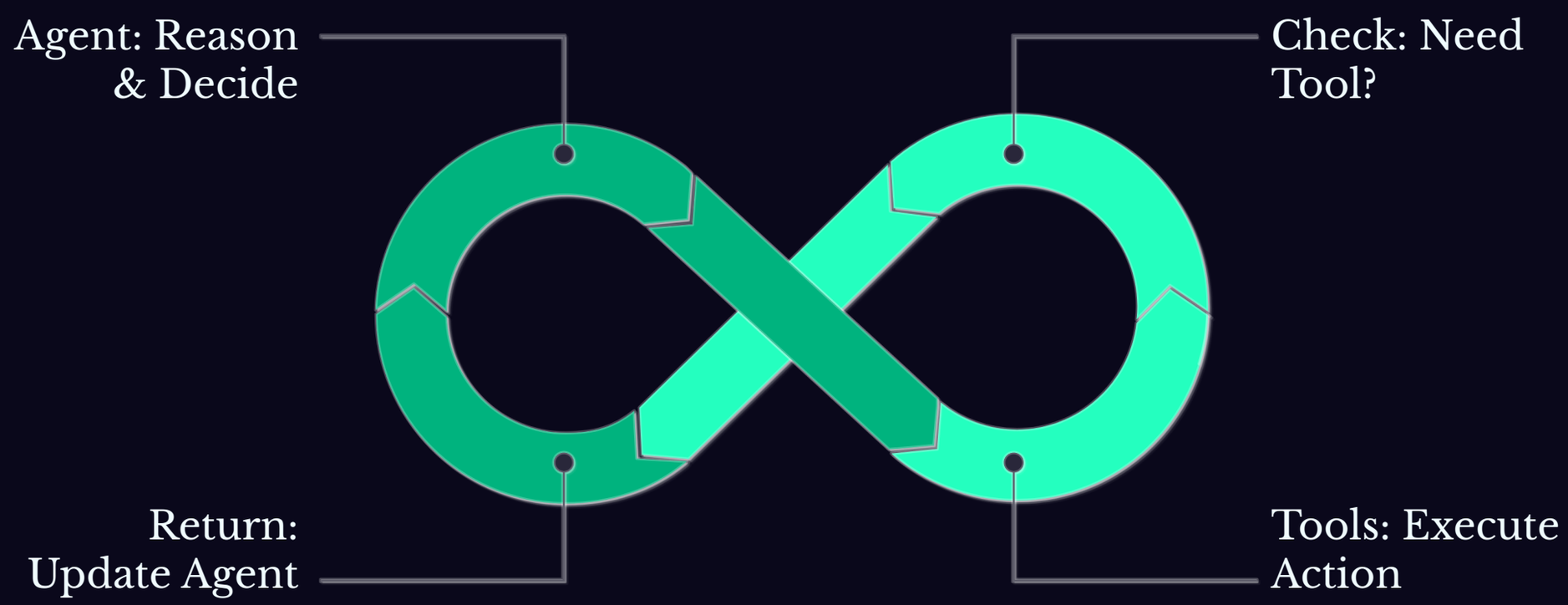
LLM reasons over conversation and decides what to do next

Tools Node

Executes whichever tool the LLM requested

Conditional Routing

Returns to agent if tool called, ends otherwise



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Interactive Demo: Student Lookup

Query: Alice's NLP Grades

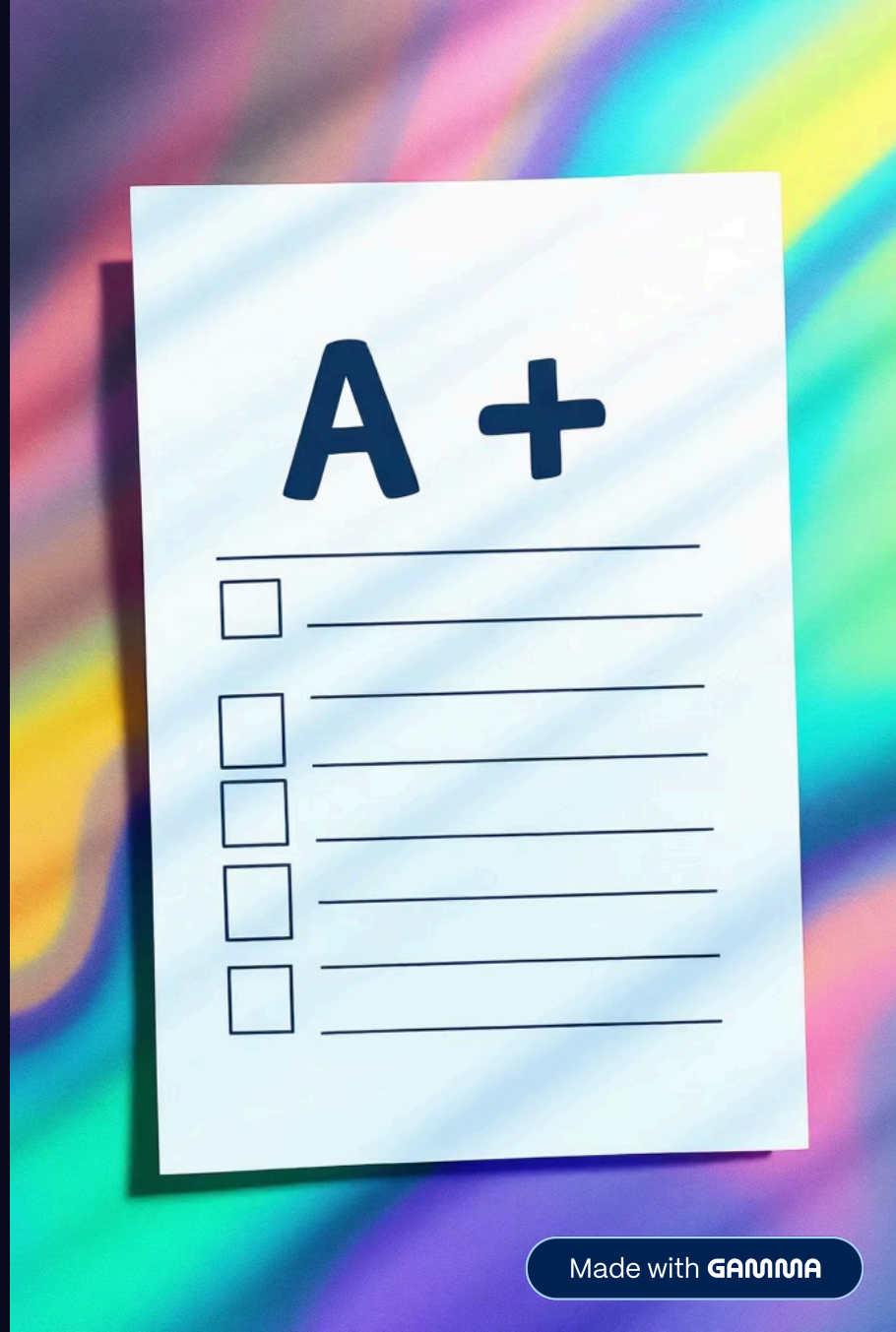
```
history = chat("What are Alice's grades in NLP?", history)
```

Tool Call: student_lookup('Alice', 'NLP')

Response

Retrieves Alice's midterm (95) and final (98) grades with weights (40% midterm, 60% final).

Database contains private data the LLM has no training access to.



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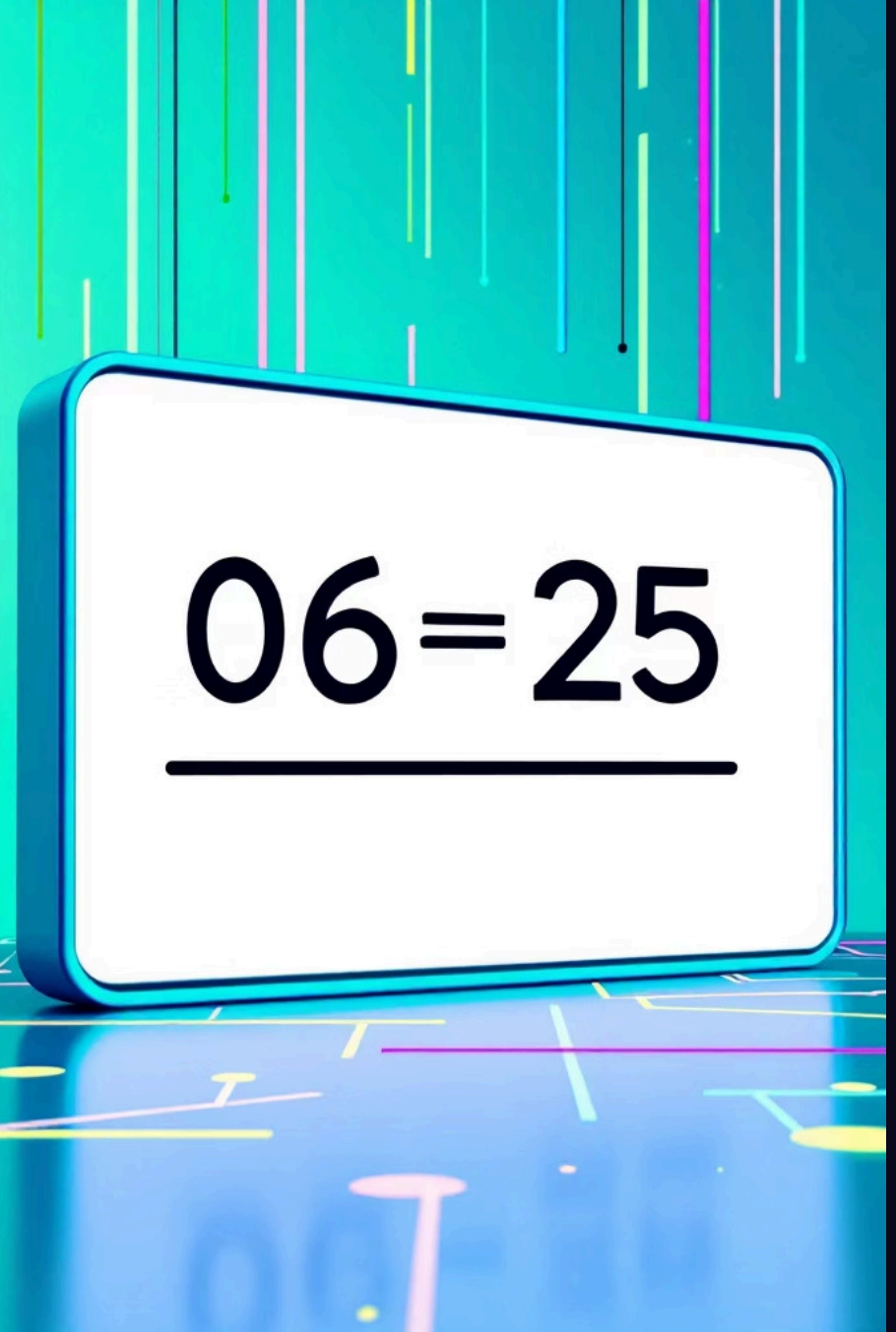

Interactive Demo: Calculator & Chaining

Bob's Math Grades

Midterm: 70 (40%) • Final: 75 (60%)

Calculate Final Grade

Uses calculator tool with weighted formula

Result: 73.0

Precise calculation: 70×0.4 + 75×0.6

The agent chains multiple tool calls automatically, looking up data then performing calculations without user intervention.

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Memory & Multi-Turn Conversations

1. **Turn 1**
   Initial greeting and capability check
2. **Turn 2-3**
   Student lookups and calculations
3. **Turn 4-5**
   Multi-tool chaining and comparisons
4. **Turn 6**
   Pure reasoning with remembered context

The `add_messages` reducer appends new messages to the list, giving the agent persistent memory across turns. The agent recalls earlier conversations without users repeating context.

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Why StudyBot Beats Plain ChatGPT

Math Accuracy

Calculator tool always returns exact answers — no hallucinated arithmetic

Private Data Access

Student lookup retrieves grades from private database the LLM has no training access to

Persistent Memory

Agent recalls earlier turns without user repeating context

Extensible Design

Add more tools (web_search, schedule_lookup) by writing new @tool functions

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